Missing Answer

1 Differentiate between: Social Networking and Social Media

Ans.1)

	Social Networking	Social Media
i.	Refers to the use of online platforms to build and maintain social relationships, communities, and networks	Refers to the use of online platforms to create, share, and exchange user-generated content
ii.	Examples of social networking sites include LinkedIn, Facebook, and Twitter	Examples of social media sites include YouTube, Instagram, and Flickr
iii.	The primary focus is on connecting with people and building relationships	The primary focus is on content creation and sharing
iv.	Users can create profiles, connect with other users, and share updates, messages, and multimedia content	Users can create, upload, and share various forms of content, such as text, images, videos, and audio
v.	Social networking sites often have privacy settings that allow users to control who can view their profiles and content	Social media sites may have privacy settings, but the emphasis is on public sharing and visibility
vi.	Social networking sites can be used for both personal and professional purposes	Social media sites are primarily used for personal purposes, but they can also be used for business and marketing
vii.	Social networking sites can facilitate the formation of new relationships and the strengthening of existing ones	Social media sites can help users discover and share new content, as well as engage with other users who have similar interests
viii.	Social networking sites can be used to build and maintain professional networks, find job opportunities, and collaborate with colleagues	Social media sites can be used to showcase personal interests, hobbies, and creativity, as well as to connect with like-minded individuals

2 Explain in detail Directed and Undirected graph with example.

Ans.2) A graph is a mathematical structure that represents a set of objects (vertices or nodes) and the relationships (edges or links) between them. Graphs can be either directed or undirected.

A directed graph, also called a digraph, is a graph where the edges have a direction, meaning that they go from one vertex to another. In a directed graph, the edges are ordered pairs of vertices, and the order matters. For example, if there is an edge from vertex A to vertex B, it does not necessarily mean that there is an edge from vertex B to vertex A. Directed graphs can be used to represent asymmetric relationships, such as a follows b on Twitter or a is a parent of b.

An undirected graph, on the other hand, is a graph where the edges do not have a direction. In an undirected graph, the edges are unordered pairs of vertices, and the order does not matter. If there is an edge between vertex A and vertex B, it means that there is also an edge between vertex B and vertex A. Undirected graphs can be used to represent symmetric relationships, such as two people are friends, or two cities are connected by a road.

Here are some examples of directed and undirected graphs:

- Directed Graph Example

In this directed graph, there are four vertices (A, B, C, and D) and four edges (A to B, B to C, C to D, and D to A). The edges have a direction, which is indicated by the arrows. For example, there is an edge from A to B, but not from B to A.

- Undirected Graph Example

In this undirected graph, there are four vertices (1, 2, 3, and 4) and four edges (1-2, 2-3, 3-4, and 4-1). The edges do not have a direction, which is indicated by the lack of arrows. For example, there is an edge between 1 and 2, which means that there is also an edge between 2 and 1.Graphs can be represented in many ways, such as adjacency matrices, adjacency lists, and edge lists. In Python, the NetworkX library provides a convenient way to create and manipulate graphs.

Here is an example of how to create a directed and an undirected graph using NetworkX:

```
import networkx as nx

# Create a directed graph
G_directed = nx.DiGraph()
G_directed.add_edge('A', 'B')
G_directed.add_edge('B', 'C')
G_directed.add_edge('C', 'D')
G_directed.add_edge('D', 'A')

# Create an undirected graph
G_undirected = nx.Graph()
G_undirected.add_edge(1, 2)
G_undirected.add_edge(2, 3)
G_undirected.add_edge(3, 4)
G_undirected.add_edge(4, 1)

# Draw the graphs
import matplotlib.pyplot as plt
nx.draw(G_directed, with_labels=True)
plt.show()
nx.draw(G_undirected, with_labels=True)
plt.show()
```

In this example, We first import the NetworkX and Matplotlib libraries. Then, we create a directed graph G_directed using the DiGraph() function and an undirected graph G_undirected using the Graph() function. We add edges to the graphs using the add_edge() method. Finally, we draw the graphs using the draw() function and show them using the show() function from Matplotlib.

3 Differentiate between Directed and Undirected graph.

Ans.3)

	Directed Graph	Undirected Graph
i.	Edges have a direction, going from one vertex to another, forming ordered pairs of vertices where the order matters	Edges do not have a direction, representing unordered pairs of vertices where the order does not matter
ii.	Suitable for representing asymmetric relationships like "follows" on social media or "parent of" relationships	Ideal for symmetric relationships such as friendships or connections between cities
iii.	Can have edges from vertex A to vertex B without an edge from B to A, indicating one-way relationships	Require an edge between vertex A and vertex B to imply an edge between B and A, showcasing mutual relationships
iv.	Used to model scenarios like email networks or citation networks where the direction of interaction matters	Commonly used for road networks or friendship networks where interactions are bidirectional and symmetric
v.	Represented by ordered pairs of vertices, emphasizing the flow or direction of relationships	Represented by unordered pairs of vertices, focusing on the existence of relationships without directionality
vi.	Examples include a network of emails sent between individuals or a network of citations between research papers	Examples include a network of roads connecting cities or a network of friendships between individuals

4 **Describe Datasets and its formats.(CSV.GML.GEFX.Pajek.Net.GraphML)**

Ans.4)

Dataset Format	Description
CSV	CSV stands for Comma Separated Values, and it is a common format for storing tabular data. In the context of graphs, a CSV file can be used to represent a graph in either edge list format or adjacency list format. In edge list format, each row of the CSV file contains two values: the source node and the target node. In adjacency list format, each row contains a node and a list of its neighbors.
GML	GML stands for Graph Modeling Language, and it is a format for storing graph data. GML files can represent both directed and undirected graphs, and they can include attributes for nodes and edges. GML files use a simple syntax that is easy to read and write.
GEXF	GEXF stands for Graph Exchange XML Format, and it is a format for exchanging graph data between different applications. GEXF files can represent both directed and undirected graphs, and they can include attributes for nodes and edges. GEXF files use a hierarchical structure based on XML, which makes them more complex than GML files but also more flexible.
Pajek	Pajek is a software package for analyzing large networks, and it has its own format for storing graph data. Pajek files can represent both directed and undirected graphs, and they can include attributes for nodes and edges. Pajek files use a simple syntax that is easy to read and write.
Net	The Net format is a simple format for storing graph data. It is used by the Pajek software package, and it can represent both directed and undirected graphs. The Net format uses a simple syntax that is easy to read and write.
GraphML	GraphML is an XML-based format for storing graph data. It is a flexible format that can represent both directed and undirected graphs, and it can include attributes for nodes and edges. GraphML files use a hierarchical structure based on XML, which makes them more complex than GML files but also more flexible.

5 Compare Selection and Social Influence.

Ans.5) Selection and social influence are two key concepts in social network analysis. Selection refers to the process of choosing who to connect with in a network, while social influence refers to the impact that connections have on individuals' behavior, attitudes, and beliefs.

Selection is often driven by homophily, the tendency for individuals to connect with others who are similar to themselves. Homophily can be based on a variety of factors, including demographic characteristics, interests, and attitudes. The selection process can lead to the formation of social clusters or communities, where individuals are more closely connected to others within their community than to those outside of it.

Social influence, on the other hand, refers to the ways in which individuals' behavior, attitudes, and beliefs are shaped by their connections in a network. Social influence can occur through a variety of mechanisms, including social comparison, conformity, and persuasion. Social influence can lead to the spread of information, behaviors, and attitudes throughout a network, and can play a key role in shaping social norms and collective behavior.

Selection and social influence are often intertwined, as the connections that individuals form through the selection process can influence their behavior and attitudes, and vice versa. For example, individuals who are connected to others who hold certain attitudes or beliefs may be more likely to adopt those attitudes or beliefs themselves. Similarly, individuals who are connected to others who engage in certain behaviors may be more likely to engage in those behaviors themselves.

Understanding the interplay between selection and social influence is crucial for understanding how social networks shape individual and collective behavior. By analyzing the patterns of connections in a network, researchers can identify the mechanisms of selection and social influence at play, and can use this information to develop strategies for promoting positive behaviors and attitudes, and for mitigating negative ones.

In summary, selection and social influence are two key concepts in social network analysis that are closely intertwined. Selection refers to the process of choosing who to connect with in a network, while social influence refers to the impact that connections have on individuals' behavior, attitudes, and beliefs. Understanding the interplay between these two concepts is crucial for understanding how social networks shape individual and collective behavior.

6 **Discuss Structural balance with all cases with the help of example.**

Ans.6) Structural balance is a concept in social network analysis that refers to the tendency of social networks to form clusters of nodes that are mutually reinforcing in their relationships. The concept is based on the idea that relationships between nodes can be either positive (friendly, cooperative) or negative (antagonistic, competitive), and that these relationships can either reinforce or undermine the relationships between other nodes in the network.

There are three main cases of structural balance:

1. **Complete balance**: In this case, all relationships between nodes in the network are positive, and there are no negative relationships. This is the most stable form of structural balance, as there are no conflicting relationships that can destabilize the network.
2. **Transitive balance**: In this case, there are both positive and negative relationships in the network, but they are arranged in such a way that they do not conflict with each other. For example, if node A has a positive relationship with node B, and node B has a positive relationship with node C, then node A and node C will also have a positive relationship. Similarly, if node A has a negative relationship with node B, and node B has a negative relationship with node C, then node A and node C will also have a positive relationship.
3. **Intransitive balance**: In this case, there are both positive and negative relationships in the network, but they are arranged in such a way that they do conflict with each other. For example, if node A has a positive relationship with node B, and node B has a positive relationship with node C, but node A has a negative relationship with node C, then the network is in a state of intransitive balance.

The concept of structural balance can be illustrated with the example of a network of three nodes (A, B, and C) and their relationships. If node A has a positive relationship with node B, and node B has a positive relationship with node C, then the network is in a state of transitive balance if node A also has a positive relationship with node C. However, if node A has a negative relationship with node C, then the network is in a state of intransitive balance. If all relationships in the network are positive, then the network is in a state of complete balance.

Structural balance is an important concept in social network analysis because it can help to explain the stability and dynamics of social networks. By understanding the patterns of relationships between nodes in a network, researchers can identify the factors that contribute to the stability or instability of the network and can develop strategies for promoting positive relationships and mitigating negative ones.

7 With the help of example Explain Schelling model of segregation.

Ans.7) The Schelling model of segregation is a theoretical framework used to explain how social and spatial segregation can emerge in a population even when individual preferences are not strongly segregationist. The model was first introduced by Thomas Schelling in 1969 and has since been used to study a wide range of phenomena, from residential segregation to social networks.

The basic idea of the Schelling model is that individuals have a preference for living in a neighborhood where a certain proportion of their neighbors are similar to them. For example, a person might prefer to live in a neighborhood where at least 30% of their neighbors are of the same race or ethnicity. If this preference is not met, the individual may choose to move to a different neighborhood.

The model is typically implemented as a computer simulation, where agents are placed on a grid and are allowed to move based on their preferences. The simulation is run iteratively, with agents updating their positions based on their preferences at each step.

There are several key insights that can be gained from the Schelling model. First, even mild preferences for living near similar others can lead to significant segregation over time. This is because as individuals move to satisfy their preferences, they can create pockets of homogeneity that attract others with similar preferences, leading to further segregation.

Second, the model suggests that segregation can be a self-reinforcing process. Once segregation has reached a certain level, it can become difficult to reverse, even if individual preferences change. This is because the pockets of homogeneity that have formed can create a feedback loop, where individuals continue to move to those areas, further reinforcing the segregation.

Third, the model highlights the importance of individual preferences in shaping social and spatial patterns. While segregation can emerge from seemingly mild preferences, it can also be disrupted by changes in those preferences. For example, if individuals become more tolerant of diversity, segregation may decrease over time.

Overall, the Schelling model of segregation is a powerful tool for understanding how social and spatial patterns can emerge and persist in a population. By simulating the behavior of individuals based on their preferences, the model can provide insights into the dynamics of segregation and the factors that contribute to it.

8 Write a note on : Diffusion in Network

Ans.8) Diffusion in networks refers to the spread of information, ideas, or innovations through a network of interconnected nodes. The process of diffusion is driven by the connections between nodes and the strength of those connections. In a network, the nodes represent individuals, organizations, or other entities, and the connections between them represent the relationships or interactions that allow for the spread of information.

The study of diffusion in networks is important for understanding how ideas, behaviors, and innovations spread through populations. For example, the spread of a new technology, the adoption of a new social norm, or the diffusion of a political idea can all be modeled and analyzed using network analysis techniques.

There are several key factors that influence the diffusion of information in networks. These include:

1. **Network structure**: The structure of the network, including the number and strength of connections between nodes, can greatly impact the speed and reach of diffusion. For example, networks with high clustering coefficients and short average path lengths are more conducive to the spread of information.
2. **Node attributes**: The attributes of individual nodes, such as their centrality, influence, or susceptibility to adoption, can also impact the diffusion process. For example, nodes with high centrality are more likely to be exposed to new information and are therefore more likely to adopt it.
3. **Tie strength**: The strength of the connections between nodes can also impact the diffusion process. Strong ties, such as close friendships or business relationships, are more likely to facilitate the spread of information than weak ties.
4. **Network dynamics**: The dynamics of the network, including the formation and dissolution of connections over time, can also impact the diffusion process. For example, networks that are dynamic and evolving may be more conducive to the spread of information than static networks.

Overall, the study of diffusion in networks is a complex and multifaceted field that requires a deep understanding of network structure, node attributes, tie strength, and network dynamics. By analyzing these factors, researchers can gain insights into the spread of information and develop strategies for promoting or inhibiting the diffusion of ideas, behaviors, and innovations.

9 Explain Epidemics in Social Network in detail.

Ans.9) Epidemics in social networks refer to the spread of diseases, ideas, or behaviors through a network of interconnected individuals. Epidemics can be modeled using network science, which allows researchers to study the structure and dynamics of social networks and how they affect the spread of diseases or behaviors.

In social networks, epidemics can spread through social contacts between individuals. For example, if one person in a social network becomes infected with a disease, they can transmit the disease to their social contacts, who can then transmit it to their own social contacts, and so on. This process can lead to the rapid spread of the disease through the network.

One important factor that affects the spread of epidemics in social networks is the structure of the network. For example, networks with high clustering coefficients and short average path lengths are more conducive to the spread of epidemics, as they allow diseases or behaviors to spread quickly through the network. On the other hand, networks with low clustering coefficients and long average path lengths are less conducive to the spread of epidemics, as they make it more difficult for diseases or behaviors to spread from one part of the network to another.

Another important factor that affects the spread of epidemics in social networks is the behavior of individuals within the network. For example, individuals who have many social contacts and engage in risky behaviors are more likely to transmit diseases or behaviors to others, while individuals who have few social contacts and engage in safe behaviors are less likely to transmit diseases or behaviors to others.

To model epidemics in social networks, researchers often use mathematical models that describe the spread of diseases or behaviors through the network. These models can be used to predict the spread of epidemics, identify key individuals or groups who are at high risk of infection, and develop strategies for controlling or preventing the spread of epidemics.

One common mathematical model used to study epidemics in social networks is the susceptible-infected-recovered (SIR) model. In the SIR model, individuals in the network are divided into three categories: susceptible (S), infected (I), and recovered (R). Susceptible individuals are those who are not yet infected but are at risk of becoming infected, infected individuals are those who are currently infected and can transmit the disease to others, and recovered individuals are those who have already been infected and have recovered and are now immune to the disease.

The SIR model describes the spread of epidemics through the network using a set of differential equations that describe the rates of change in the number of susceptible, infected, and recovered individuals over time. These equations can be used to predict the spread of epidemics and identify key individuals or groups who are at high risk of infection.

In summary, epidemics in social networks refer to the spread of diseases, ideas, or behaviors through a network of interconnected individuals. The spread of epidemics in social networks is affected by the structure of the network and the behavior of individuals within the network. Mathematical models, such as the SIR model, can be used to predict the spread of epidemics and identify key individuals or groups who are at high risk of infection.

10 Compare SIR and SIS spreading models.

Ans.10) SIR and SIS are two common models used to describe the spread of diseases in a population. Here is a comparison of the two models in tabular form:

	SIR Model	SIS Model
State Variables	Susceptible (S), Infected (I), Recovered (R)	Susceptible (S), Infected (I)
Description	Individuals can transition from susceptible to infected to recovered state	Individuals can transition back and forth between susceptible and infected state
Final State	Some individuals remain in the recovered state	All individuals eventually return to the susceptible state
Applications	Used to model diseases with immunity, such as measles or chickenpox	Used to model diseases without immunity, such as the common cold or flu
Example	Measles outbreak in a school	Flu outbreak in a office

The SIR model is used to describe diseases where individuals can become immune after recovering from the disease. In contrast, the SIS model is used to describe diseases where individuals do not become immune after recovering from the disease and can become infected again.

The SIR model is often used to model diseases with a high severity, such as measles or chickenpox, where individuals who recover from the disease have lifelong immunity. In contrast, the SIS model is often used to model diseases with a low severity, such as the common cold or flu, where individuals who recover from the disease can become infected again.

The SIR model is useful for understanding the spread of diseases in a population and identifying strategies to control the spread of the disease. For example, by identifying the basic reproductive number (R0) of the disease, public health officials can determine the proportion of the population that needs to be immunized to control the spread of the disease.

The SIS model is useful for understanding the dynamics of diseases in a population and identifying strategies to control the spread of the disease. For example, by identifying the basic reproductive number (R0) of the disease, public health officials can determine the proportion of the population that needs to be treated to control the spread of the disease.

In summary, the SIR and SIS models are two common models used to describe the spread of diseases in a population. The SIR model is used to model diseases with immunity, while the SIS model is used to model diseases without immunity. Both models are useful for understanding the spread of diseases in a population and identifying strategies to control the spread of the disease.

11 Describe following Social Network dataset formats:

1. CSV 2. GML 3. Pajeck 4. GraphML 5. GEFX.

Ans.11) Social Network Dataset Formats:

1. **CSV (Comma Separated Values)**: This format is used to represent networks in either edge list or adjacency list format. Each row in the CSV file contains two values: the source node and the target node, indicating an edge between them. The simplicity of this format makes it widely used for network data representation.

2. **GML (Graph Modeling Language)**: This is a commonly used format for network datasets due to its flexibility in labeling or assigning attributes to nodes and edges. The format is not complex and allows for easy manipulation of network data. GML files start with the "graph" keyword, followed by square brackets containing node and edge details.

3. **Pajek**: This format is used for network datasets and is supported by various network analysis tools. Pajek files contain node and edge details, with nodes listed first followed by edge details. The format is simple and easy to understand, making it a popular choice for network data representation.

4. **GraphML**: This format is based on XML and provides a hierarchical structure for storing network data. It allows for the assignment of attributes to nodes and edges using various tags. The structure of GraphML files can be complex, but they offer greater flexibility in storing network data compared to other formats.

5. **GEFX (Graph Exchange XML Format)**: This format was created by Gephi, an open-source software for visualizing and analyzing social networks. GEXF files are composed of various XML tags and are similar to GraphML in structure. They are used to store network data with attributes for nodes and edges.

These formats are widely used for storing and sharing network data, and each has its strengths and weaknesses. The choice of format depends on the specific needs of the network analysis task at hand.